
Design and Implementation of a Dual-Evaporator Transcritical High Temperature Heat Pump

Research Project

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1 Context

According to the Global Energy Review 2025 report published in March 2025 by the IEA [1], the global CO₂ emissions have increased by 0.8% in 2024, reaching an all-time high of 37.8 Gt CO₂ a year. In addition, the report highlights that the energy demand grew by 2.2% in 2024. This increase in CO₂ emissions is totally inconsistent with the goals of the Paris Agreement, which aims to abate the CO₂ emissions by 43% by 2030 and to reach net-zero emissions by 2050. However, the lower increase of CO₂ emissions compared to the energy demand growth is an encouraging trend, as it suggests a relative decoupling of emissions from energy consumption.

In [4], we can read that the waste heat potential in the EU-27 and UK reached 221.32 TWh/year in 2021. In comparison, the demand for less than 140°C heat in Germany was 612 PJ, or 170 TWh in 2012 [3]. These numbers justify the interest from companies/researchers to design and study machines capable of recovering this waste energy.

In this work, we will focus on the use of high temperature heat pumps (HTHP's) for recovering waste heat from industrial processes. In [3], the authors provide a list of manufacturers and universities that have already been working on the design of several machines for at least one decade. The already existing machines cover a large spectrum of applications, going from small machines of a few thousands watts to big ones of more than one megawatt. However, new design ideas can still be studied in order to improve the current recovery systems.

The main purpose of this work is to design the thermodynamical cycle of a transcritical dual-evaporator HTHP, build it and test its performances in laboratory conditions. No similar applications were found in the literature. Besides, the machine build in the frame of this master thesis should be used in the future for an academic purpose.

2 Literature review

As discussed in the previous section, no publications specifically addressing dual-evaporator transcritical HTHPs were identified. Nevertheless, a brief review of the current state-of-the-art technology for HTHPs was carried out. This review has led to the identification of several reasons why the topic may be particularly relevant for certain applications and therefore is worth further investigation.

First, regarding the dual-evaporator configuration: it has been shown in [5] that using multiple evaporators offers significant potential for waste heat recovery applications. In the case study considered, notable improvements in both the Coefficient of Performance (COP) and Volumetric Heating Capacity (VHC) were observed when two evaporators were employed instead of one, once a significant portion of the available heat was present at the medium temperature (MT) level (see Figure 1).

This proportion of heat available at the medium temperature level relative to the total heat input is quantified by the heat source ratio parameter (β), first introduced in [2]:

$$\beta \triangleq \frac{\dot{Q}_{MT}}{\dot{Q}_{MT} + \dot{Q}_{LT}}$$

where $\dot{Q}_{MT}$ and $\dot{Q}_{LT}$ represent the thermal powers captured at the medium and low temperature evaporators, respectively.

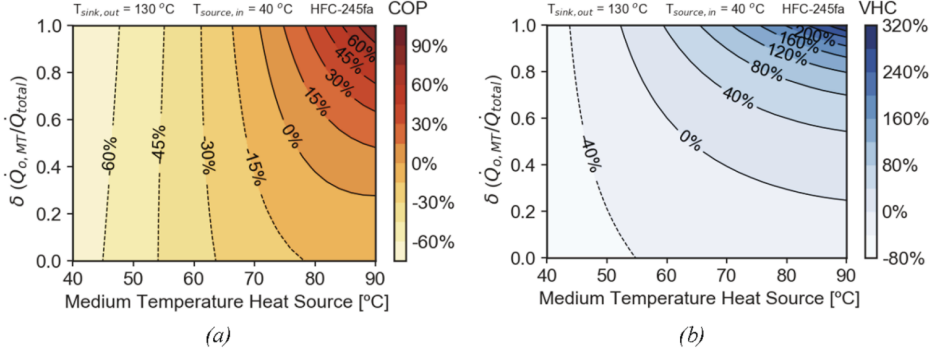


Fig. 15. Performance parameters evaluation of the TS Booster cycle compared to the reference TS Economizer, operating with HFC-245fa: a) COP and b) VHC.

Figure 1: COP and VHC increase significantly when using two evaporators instead of one, once a substantial portion of the total captured heat is available at the medium-temperature source. Figure taken from [5].

While the benefit of a higher COP is not to be demonstrated, the VHC is also of particular interest. As explained in [3], the VHC is directly related to the refrigerant charge and size of the compressor: a higher VHC allows for a smaller refrigerant charge and a more compact compressor, thereby reducing system costs. Consequently, both the COP and the VHC are important factors in designing efficient and cost-effective heat pumps.

Next, regarding the transcritical aspect of the cycle: in [6], the authors demonstrate that using a transcritical cycle instead of a subcritical one can lead to significant improvements in COP when the temperature glide at the heat sink is high. According to their study, the ideal heat sink temperature glide is around 60 K. This COP increase is attributed to a lower temperature mismatch during heat transfer at the heat sink (see Figure 2).

The improved temperature matching results from the properties of a supercritical fluid. Since heat transfer occurs above the critical pressure, the fluid does not undergo a phase change. Therefore, its temperature can vary significantly during heat transfer, allowing for better alignment with the heat sink fluid temperature and reduced exergy losses. The authors also show that transcritical cycles generally achieve higher VHC values than subcritical cycles.

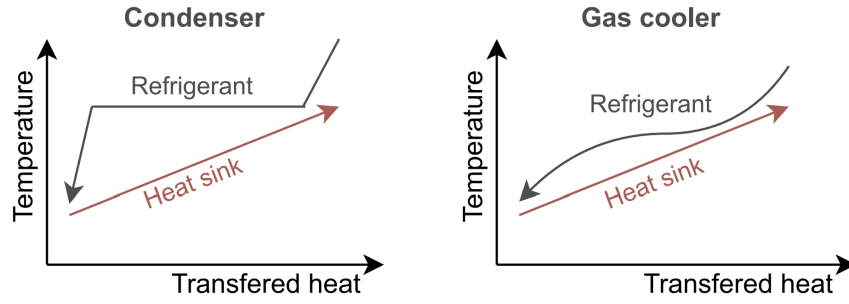


Fig. 1. Temperature profile during heat rejection for subcritical operation (condenser) and transcritical or supercritical operation (gas cooler).

Figure 2: The area between the two curves represents the exergy destruction due to heat transfer. This area is considerably smaller in the transcritical case, resulting in lower irreversibilities and a potentially higher COP. Figure taken from [6].

Finally, on the high temperature. First, high temperature heat pumps are classified by a heat sink temperature higher than 100°C [3]. Classical heat pumps are an already well established technology for domestic applications.

To be completed.

In conclusion, a dual-evaporator transcritical high-temperature heat pump has the potential to outperform current HTHPs on the market, particularly in applications where heat is available at two temperature levels (e.g., waste heat recovery) and where the temperature glide at the heat sink is high. The aim of this Master's thesis is to investigate this concept, contributing to the broader effort to advance high-temperature heat pumps and decarbonize industrial heat production.

3 Research question

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4 Project description

In order to answer to the previous research question, we distinguish six steps in this master thesis. These steps can be observed in the Figure ...

Preliminary cycle design

In this first part, we will numerically modeled the thermodynamical design. Each student will work simultaneously on different blocks, determined in advance. The analysis of the cycle will include an energy and exergy analysis to ensure optimal performances. The numerical model will be written in Python and the module Coolprop will be used to access numerical thermodynamics tables.

Final cycle design

Assembling all the different blocks will be the purpose of the second part. Beside connecting the blocks, an optimization process will be developped to determine the optimal parameters in nominal conditions. In addition, more physical aspects will be added to the model such as head losses computations and heat transfer modeling.

Control

One main advantage of this machine will be its capacity of simulating different conditions thanks to the dual evaporator. A control logic must therefore be implemented. This part is critical because we'll have to think about the nature and the location of the sensors. A student in industrial engineering will bring his expertise and will start to collaborate with us.

Machine design

In this part, the other student will be in charge of the design of the machine. It will contain material/components selection, heat exchangers and insulation design and the elaboration of precise technical draw. That will be our task to correct the final cycle design (head losses, heat exchange, ...). We will also collaborate with Frank, the technician that will be responsible of the buiding of the phase. We surely think that its expertise will be needed for many design uncertainties.

Building

The building phase will be ensured by Frank and other technicians of the CREDEM.

Testing

Finally, the last part will be dedicated to the evaluation of the machine. Several tests will take place to ensure the good operation of the machine and to collect data in order to compare them to theoretical values. We don't think it will possible to have a correction phase in case of underperforming.

5 Share of workload and responsibilities

To be completed by FG (1 page).

The workload of this Master's thesis project will be shared equally. To maximize productivity, we plan to work in parallel on different subtasks. This productivity will be crucial since, as will be discussed in the next section, the sizing phase of the different components should be completed before the end of the first semester to be able to have the components on time for the prototyping phase to start early in the second semester. Nonetheless, we will collaborate closely on strategic subtasks, important design decisions and provide mutual support when required.

Weekly peer meetings will be organized throughout the whole duration of the project. These meetings will be used to discuss the progress of the project, share knowledge, and make important decisions. In addition, we will meet our mentors on a regular basis (ideally weekly) to present important results, get feedback and discuss the main challenges. During the

During the prototyping phase, meetings with the workshop technician will be held on a regular basis too.

Additionally, our progress will be presented in a more formal way during monthly meetings with our supervisor.

6 Roadmap and planning

To be completed by HW (2 pages).

References

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